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Logic and Approach for solving pattern programm Questions.

Pattern programming questions are based on observing how characters, number or symbols are arranged row by row, instead of memorizing sollutions, it is important to understand the logic behind the pattern.

1. Identify the number of rows

The first step is to determine how many rows the pattern contains. The number of rows usually controls the outer loop.

2- Analyze each row

observe what changes from one row to next. check:
→ number of spaces
→ number of star or symbol
→ number sequence.

3. Divide the pattern into parts

most pattern consist of
→ leading spaces
→ symbols (stars, numbers, letters)
→ trailing spaces.

Analyze each part seprately.

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7. Create a table before coding

Before writing code, create a table showing the values for each row.

Row	spaces	stars
1	3	1
2	2	3
3	1	5
4	0	7

8. Determine whether the pattern is increasing or Decreasing

observe whether the number of symbols

→ Increases in each row
→ Decreases in each row
→ Increases and then decreases.

This observation helps in designing loop conditions.

9. Write the algorithm

A general algorithm for pattern problem is

1. Read the number of rows
2. Run an outer loop for each row
3. Print spaces if required
4. Print stars, numbers, or characters according to the pattern
5. Move to the next line
6. Repeat until all rows are printed.

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4. Find the relationship between Row number and output.

For every row, determine a formulla for

- Spaces
- Stars
- Numbers

Common relationships include:

- Increasing count: row number (i)
- Decreasing count: $n-i$
- Pyramid stars: $2 \times i - 1$

5. Use nested loops

Pattern problems are genreally solved using nested loops

→ outer loops control the row.
→ Inner loop control spaces, stars, number or other characters.

6. Check for symmetry

Patterns such as diamond, butterflies, and hourglasses are often symmetric. Then pattern can usually be divided into:

- Upper half
- lower half.

Each half Can be Solved seprately.

Topics to be Covered:

1. **Star Patterns**

1. Square

```
*****
*****
*****
*****
*****
```

2. Hollow Square

```
*****
*   *
*   *
*   *
*****
```

3. Triangle

```
*
**
***
****
*****
```

4. Inverted Triangle

```
*****
 ****
  ***
   **
    *
```

5. Pyramid

```
    *
   ***
  *****
 *******
*********
```

6. Diamond

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

7. Half Diamond

```
*
**
***
****
*****
****
***
**
*
```

8. Half Diamond Inverted

```
*****
 ****
  ***
   **
    *
```

assignement_Q1.cpp

assignement_Q1.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for(int i=0;i<5;i++)
    {
        for (int j=0;j<5;j++)
        {
            cout<<"*";
        }
        cout<<endl;
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q1.cpp -o assignement_Q1 } ; if ($?) { .\assignement_Q1 }
*****
*****
*****
*****
*****
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Code

Code

assignement_Q10.cpp | assignement_Q11.cpp | assignement_Q12.cpp | assignement_Q13.cpp

assignement_Q13.cpp > main()

```cpp
#include<iostream>
using namespace std;

int main()
{
    int n = 5;

    for(int i=1; i<=n; i++)
    {
        for(int j=1; j<=n; j++)
        {
            if(i==1 || i==n || j==1 || j==n)
                cout<<"* ";
            else
                cout<<"  ";
        }
        cout<<endl;
    }

    return 0;
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
* * * * *
*       *
*       *
*       *
* * * * *
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

assignement_Q1.cpp | assignement_Q2.cpp | assignement_Q3.cpp | assignement_Q4.cpp | assignement_Q5.cpp

assignement_Q5.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=1;i<=5;i++)
    {
        for (int j=1;j<=i;j++)
        {
            cout<<"*";
        }
        cout<<endl;
    }
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q5.cpp -o assignement_Q5 } ; if ($?) { .\assignement_Q5 }
*
**
***
****
*****
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 7, Col 21   Spaces: 4   UTF-8   CRLF   { } C++   Go Live   Win32

signement_Q1.cpp | assignement_Q2.cpp | assignement_Q3.cpp | assignement_Q4.cpp | assignement_Q5.cpp | assignement_Q6.cpp | assignement_Q7.cpp

assignement_Q7.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=5;i>0;i--)
    {
        //space printing
        for (int k = 0; k < 5- i; k++)
        {
            cout << " ";
        }
        //star printing
        for (int j=0;j<i+(i-1);j++)
        {
            cout<<"*";
        }
        cout<<endl;

    }

}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q7.cpp -o assignement_Q7 } ; if ($?) { .\assignement_Q7 }
*********
 *******
  *****
   ***
    *
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 15, Col 23   Spaces: 4   UTF-8   CRLF   { } C++   Go Live   Win32

assignement_Q1.cpp | assignement_Q2.cpp | assignement_Q3.cpp | assignement_Q4.cpp | assignement_Q5.cpp | assignement_Q6.cpp

assignement_Q6.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=0;i<5;i++)
    {
        //space 
        for (int j=0;j<5-i-1;j++)
        {
            cout<<" ";
        }
        //star
        for (int k=0;k<i+(i+1);k++)
        {
            cout<<"*";
        }
        //space
        for (int l=0;l<5-i-1;l++)
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q6.cpp -o assignement_Q6 } ; if ($?) { .\assignement_Q6 }
    *
   ***
  *****
 *******
*********
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 7, Col 17   Spaces: 4   UTF-8   CRLF   { } C++   Go Live   Win32

ignement_Q2.cpp | assignement_Q3.cpp | assignement_Q4.cpp | assignement_Q5.cpp | assignement_Q6.cpp | assignement_Q7.cpp | assignement_Q8.cpp

assignement_Q8.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=0;i<5;i++)
    {
        //space
        for (int j=0;j<5-i-1;j++)
        {
            cout<<" ";
        }
        //star
        for (int k=0;k<i+(i+1);k++)
        {
            cout<<"*";
        }
        cout<<endl;


    }
    for (int i=4;i>0;i--)
    {
        //space printing
        for (int k = 0; k < 5- i; k++)
        {
            cout << " ";
        }
        //star printing
        for (int j=0;j<i+(i-1);j++)
        {
            cout<<"*";
        }
        cout<<endl;
    }
}
```

Ln 7, Col 17 Spaces: 4 UTF-8 CRLF { } C++ Go Live Win32

PROE Focus folder in explorer (ctrl + click) TERMINAL PORTS Code

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q8.cpp -o assignement_Q8 } ; if ($?) { .\assignement_Q8 }
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 7, Col 17 Spaces: 4 UTF-8 CRLF { } C++ Go Live Win32

assignement_Q9.cpp   Untitled-1

assignement_Q9.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for(int k=5;k>=1;k--)
    {
        for(int l=k;l<=5;l++)
        {
            cout<<"*";
        }
        cout<<endl;
    }




    for (int i=1;i<5;i++)
    {
        for(int j=i;j<5;j++)
        {
            cout<<"*";
        }
        cout<<endl;
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q9.cpp -o assignement_Q9 } ; if ($?) { .\assig
nement_Q9 }
*
**
***
****
*****
****
***
**
*
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 9, Col 25 Spaces: 4 UTF-8 CRLF { } C++ Go Live Win32

signement_Q1.cpp | assignement_Q2.cpp | assignement_Q3.cpp | assignement_Q4.cpp | assignement_Q5.cpp | assignement_Q6.cpp | assignement_Q7.cpp

assignement_Q7.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=5;i>0;i--)
    {
        //space printing
        for (int k = 0; k < 5- i; k++)
        {
            cout << " ";
        }
        //star printing
        for (int j=0;j<i+(i-1);j++)
        {
            cout<<"*";
        }
        cout<<endl;

    }

}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q7.cpp -o assignement_Q7 } ; if ($?) { .\assignement_Q7 }
*********
 *******
  *****
   ***
    *
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 15, Col 23 Spaces: 4 UTF-8 CRLF {} C++ Go Live Win32

2. Number Patterns

1. Square	2. Incrementing	3. Right Triangle	4. Inverted Diamond
1 1 1 1 1	1 2 3 4 5	1	3 3 3 3 3
1 1 1 1 1	1 2 3 4 5	1 2	4 4 4 4
1 1 1 1 1	1 2 3 4 5	1 2 3	5 5 5
1 1 1 1 1	1 2 3 4 5	1 2 3 4	6 6
1 1 1 1 1	1 2 3 4 5	1 2 3 4 5	7

5. Sandwich Pattern

```
3 3 3 3 3
 4 4 4
   5
 4 4 4
3 3 3 3 3
```

6. Incrementing Diamond Pattern

```
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```

assignement_Q1.cpp   assignement_Q2.cpp

assignement_Q2.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=0;i<5;i++)
    {
        for (int j=0;j<5;j++)
        {
            cout<<"1";
        }
        cout<<endl;
    }
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q2.cpp -o assignement_Q2 } ; if ($?) { .\assignement_Q2 }
11111
11111
11111
11111
11111
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

assignement_Q1.cpp | assignement_Q2.cpp | assignement_Q3.cpp | assignement_Q4.cpp

assignement_Q3.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=1;i<=5;i++)
    {
        for (int j=1;j<=5;j++)
        {
            cout<<j;
        }
        cout<<endl;
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q3.cpp -o assignement_Q3 } ; if ($?) { .\assig
nement_Q3 }
12345
12345
12345
12345
12345
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 13, Col 1 Spaces: 4 UTF-8 CRLF { } C++ Go Live Win32

assignement_Q1.cpp | assignement_Q2.cpp | assignement_Q3.cpp | assignement_Q4.cpp

assignement_Q4.cpp > main()

```cpp
#include<iostream>
using namespace std;
int main()
{
    for (int i=1;i<=5;i++)
    {
        for (int j=1;j<=i;j++)
        {
            cout<<j;
        }
        cout<<endl;
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q4.cpp -o assignement_Q4 } ; if ($?) { .\assignement_Q4 }
1
12
123
1234
12345
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Ln 7, Col 26 Spaces: 4 UTF-8 CRLF { } C++ Go Live Win32

assignement_Q10.cpp

assignement_Q10.cpp > main()

```cpp
#include<iostream>
using namespace std;

int main()
{
    for(int i=3;i<=7;i++)
    {
        // spaces
        for(int k=0;k<i-3;k++)
        {
            cout<<" ";
        }

        // numbers
        for(int j=0;j<8-i;j++)
        {
            cout<<i<<" ";
        }

        cout<<endl;
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q10.cpp -o assignement_Q10 } ; if ($?) { .\assignement_Q10 }
3 3 3 3 3
 4 4 4 4
  5 5 5
   6 6
    7
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

Spaces: 4 UTF-8 { } C++ Go Live Win32

assignement_Q10.cpp assignement_Q11.cpp #include<iostream> Unt

```
#include<iostream>
using namespace std;

int main()
{
    // upper half
    for(int i=1;i<=3;i++)
    {
        for(int s=1;s<i;s++)
            cout<<" ";

        for(int j=1;j<=5-2*(i-1);j++)
            cout<<i+2<<" ";

        cout<<endl;
    }

    // lower half
    for(int i=2;i>=1;i--)
    {
        for(int s=1;s<i;s++)
            cout<<" ";

        for(int j=1;j<=5-2*(i-1);j++)
            cout<<i+2<<" ";

        cout<<endl;
    }

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) {
 .\tempCodeRunnerFile }
3 3 3 3 3
 4 4 4
  5
 4 4 4
3 3 3 3 3
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```

assignement_Q10.cpp   assignement_Q11.cpp

assignement_Q11.cpp > main()

```cpp
#include<iostream>
using namespace std;

int main()
{

    for(int i=1;i<=5;i++)
    {
        for(int j=1;j<=i;j++)
        {
            cout << i+2 << " ";
        }
        cout << endl;
    }


    for(int i=4;i>=1;i--)
    {
        for(int j=1;j<=i;j++)
        {
            cout << i+2 << " ";
        }
        cout << endl;
    }

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\Desktop\C++\DSA\PATTERN> cd "c:\Users\dell\Desktop\C++\DSA\PATTERN\" ; if ($?) { g++ assignement_Q11.cpp -o assignement_Q11 } ; if ($?) { .\assignement_Q11 }
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
PS C:\Users\dell\Desktop\C++\DSA\PATTERN>
```